

Simple document

I'm an R Markdown document!

Section 1

Here's a **code chunk** that samples from a *normal distribution*:

```
samp = rnorm(100)
length(samp)
```

```
## [1] 100
```

Section 2

I can take the mean of the sample, too! The mean is -0.1485419.

Section 3: a tibble

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2     4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

## # A tibble: 6 x 2
##       x         y
##   <dbl> <dbl>
## 1  0.0119  1.41
## 2 -0.00404 0.809
## 3  0.0334  1.08
## 4 -0.0127  1.53
## 5 -0.0807 -0.425
## 6  0.0726  1.87
```

Section 5: Learning Assessment

Write a named code chunk that creates a dataframe comprised of: a numeric variable containing a random sample of size 500 from a normal variable with mean 1; a logical vector indicating whether each sampled value is greater than zero; and a numeric vector containing the absolute value of each element. Then, produce a histogram of the absolute value variable just created. Add an inline summary giving the median value rounded to two decimal places. What happens if you set `eval = FALSE` to the code chunk? What about `echo = FALSE`?

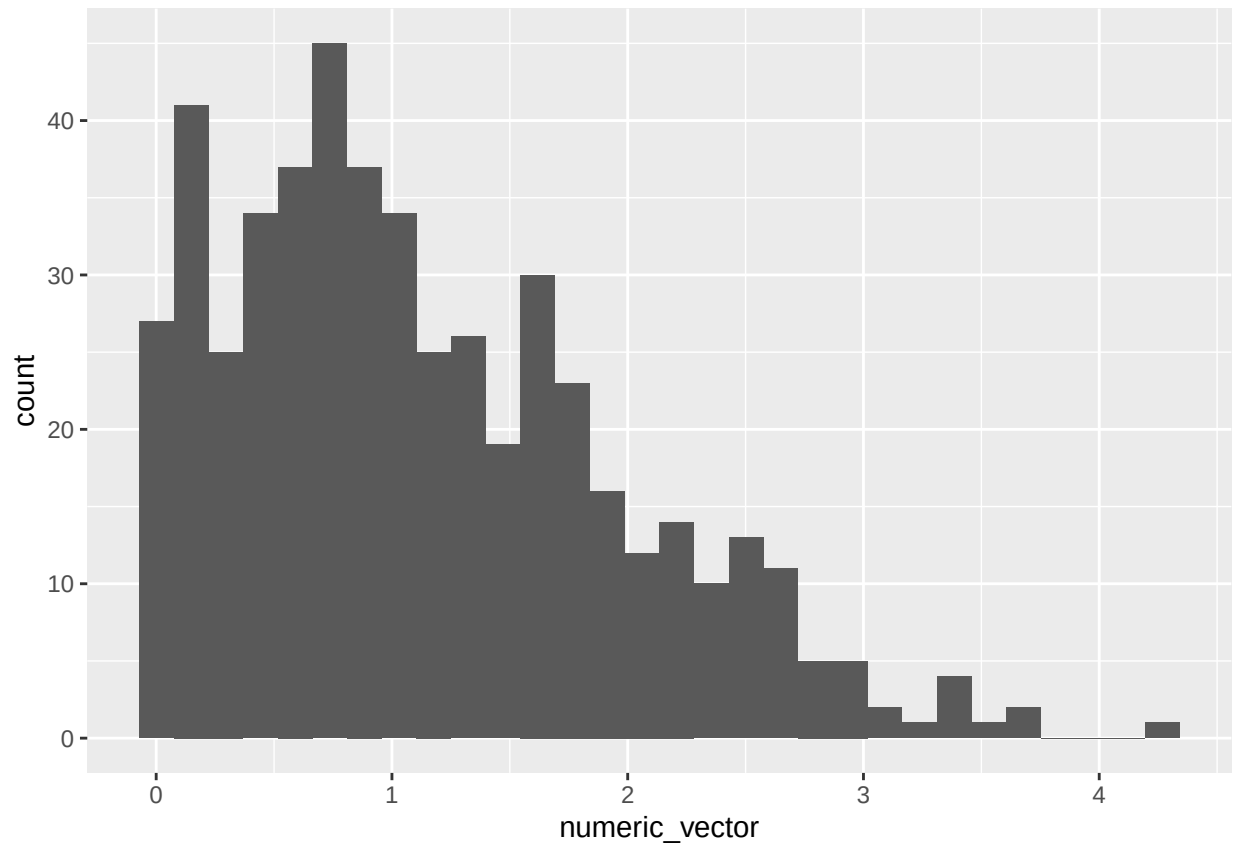
Solution

```
set.seed(0)

data_frame <-
  tibble(
    numeric_var = rnorm(500, mean = 1),
    logical_vector = case_when(
      numeric_var >= 0 ~ 1,
      numeric_var < 0 ~ 0),
    log_var = numeric_var > 0,
    numeric_vector = abs(numeric_var)
  )

data_frame |>
  ggplot() +
  geom_histogram(mapping = aes(x = numeric_vector))
```

```
## 'stat_bin()' using 'bins = 30'. Pick better value 'binwidth'.
```



```
round(median(data_frame$numeric_vector), digits = 2)
```

```
## [1] 0.99
```

```
# If eval = FALSE, then the block of code does not run  
# If echo = FALSE, then block of code does not appear in R Markdown file
```

The median is 0.99
The median is 0.99

Section 6 Formatting

Text formatting

italic or *italic* **bold** or **bold** code superscript² and subscript₂

Headings

1st Level Header

2nd Level Header

3rd Level Header

Lists

- Bulleted list item 1
 - Item 2
 - Item 2a
 - Item 2b
1. Numbered list item 1
 2. Item 2. The numbers are incremented automatically in the output.

Tables

First Header	Second Header
Content Cell	Content Cell
Content Cell	Content Cell

Section 7: Learning Assessment 3

After the previous code chunk, write a bullet list given the mean, median, and standard deviation of the original random sample.

- The median is 0.99.
- The mean is 1.14.
- The standard deviation is 0.82.

This plot shows the distribution of the absolute value.